DR IMAN NABIL HISTOLOGY

Cell organelles 1

Dr.Iman Nabil

DrImanNabil

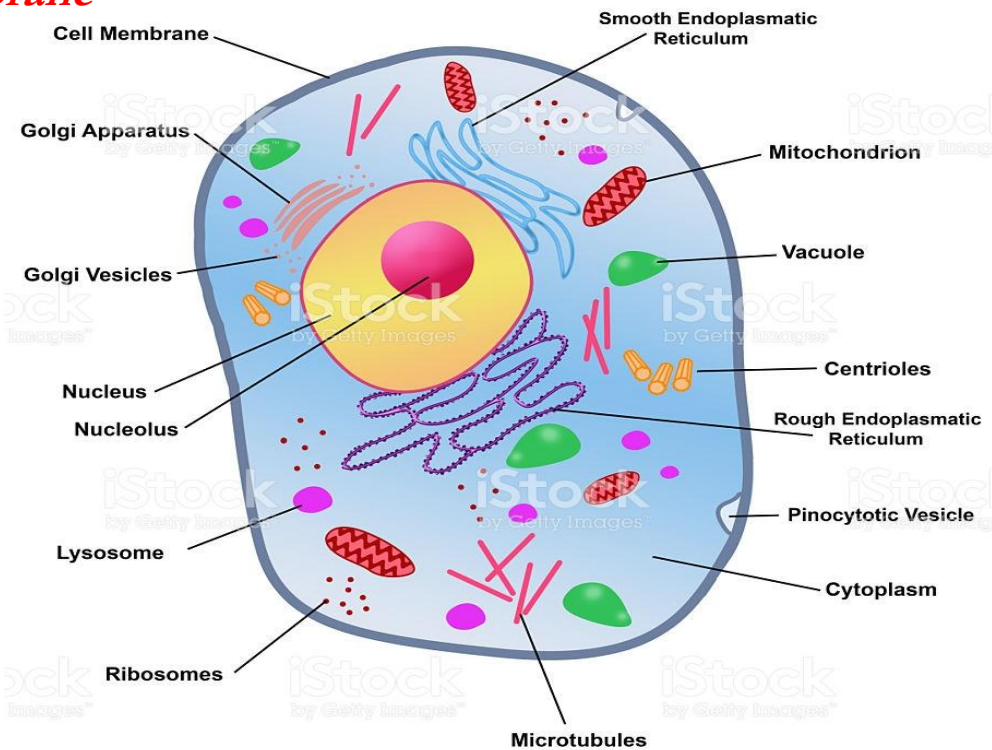

Why?
tell me
why!



ANIMAL CELL STRUCTURE

Cell membrane

Inclusions



Organelles

Cytoskeleton

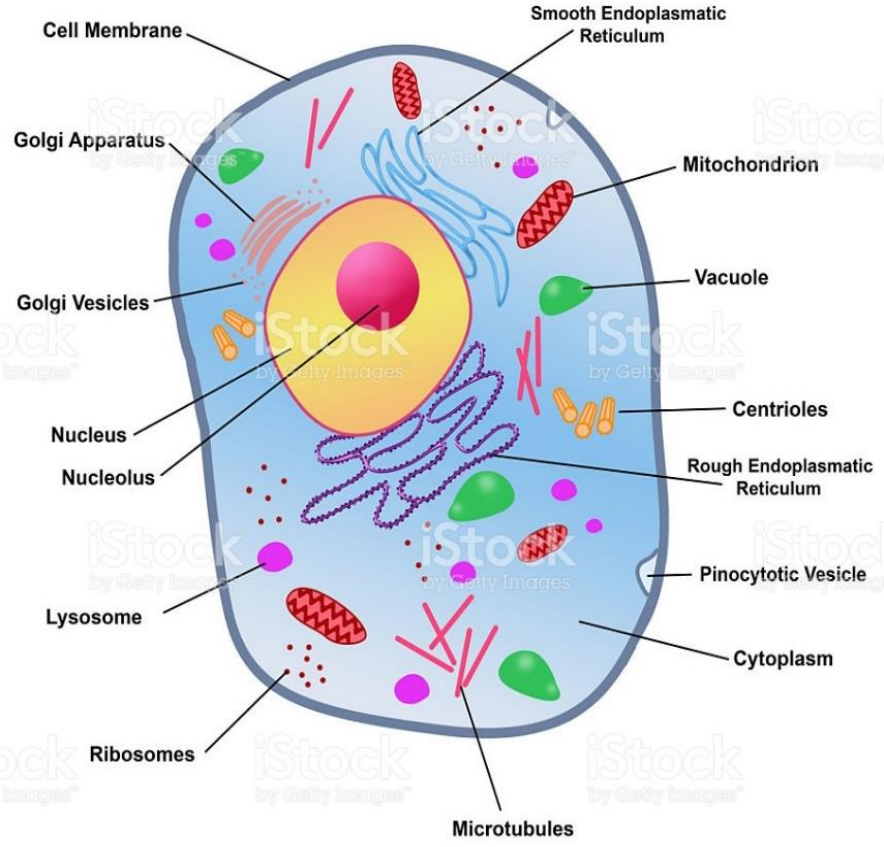
Cell organelles

- **Definition**: metabolically-active structures carrying out specific essential functions.
- **Types**:

1- Membranous organelles:

nucleus, endoplasmic reticulum, Golgi apparatus, transport vesicles, endosomes, lysosomes, mitochondria and peroxisomes.

2- Non-membranous organelles: ribosomes, centrosome and the cytoskeleton.



Cell organelles I

1- Ribosomes: structure& function.

2- Endoplasmic reticulum: Types, structure& functions.

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LM

EM

Functions

Haematoxylin and Eosin (H&E)

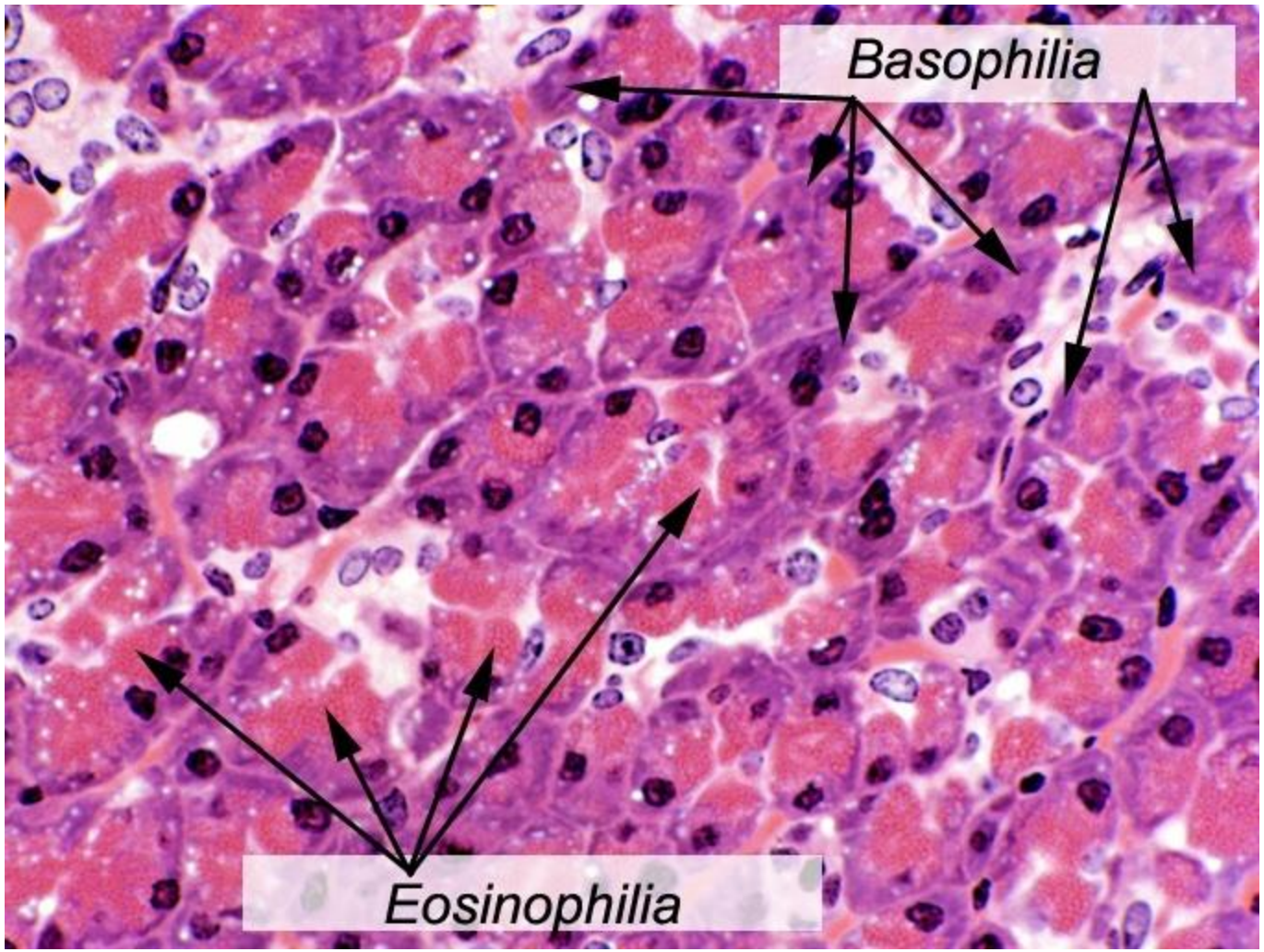
- **Haematoxylin is a basic violet stain.**

- **Eosin is an acidic pink stain.**

- **Basophilic structure= A structure that has affinity to stain with basic dyes = acidic in nature…….. So it stains violet with haematoxylin**

- **Acidophilic or eosinophilic structure= A structure that has affinity to stain with acidic dyes = basic in nature……So it stains pink with eosin**

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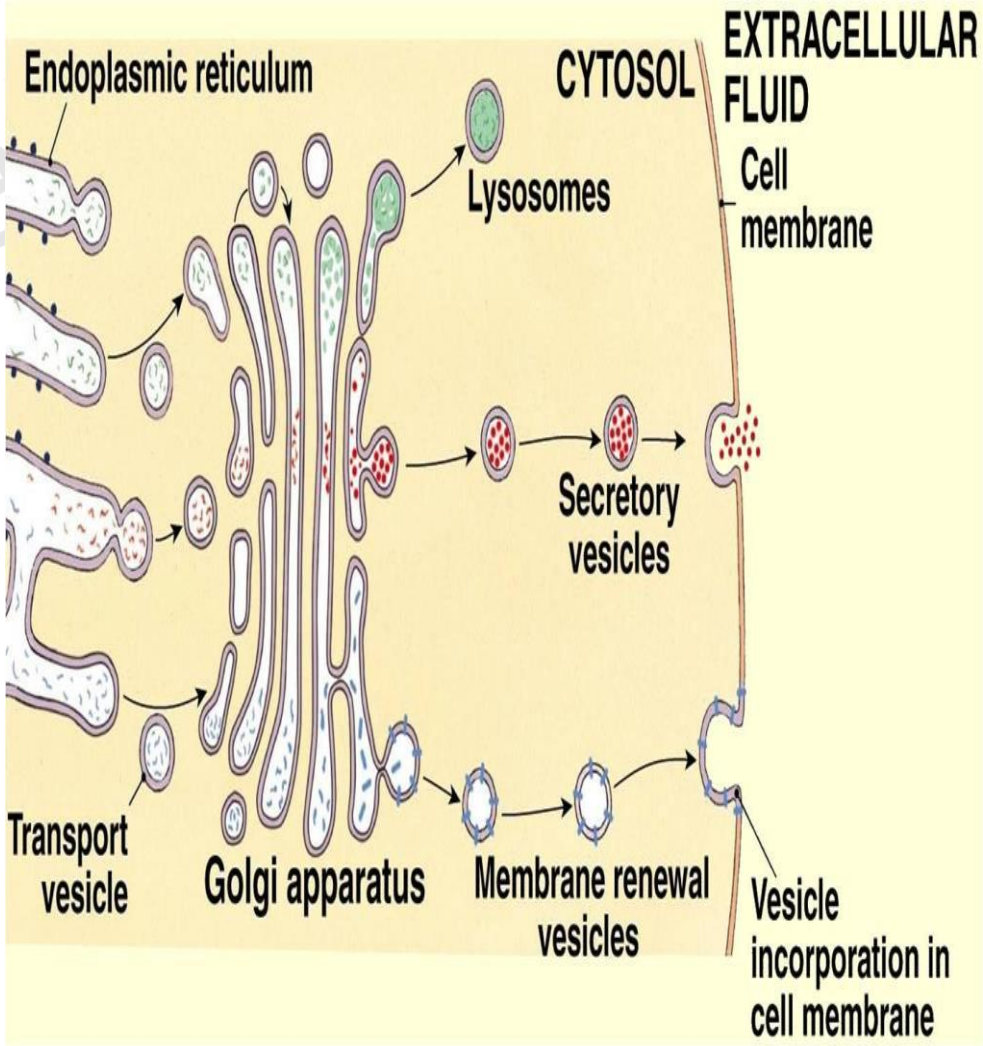

Basophilia
Eosinophilia

Cytoplasmic Organelles that participate in the process of protein synthesis

1- Ribosomes (factories)

2- Rough endoplasmic reticulum (modification& transport)

3- Golgi apparatus (chemical modification,package,sorting& targeting)



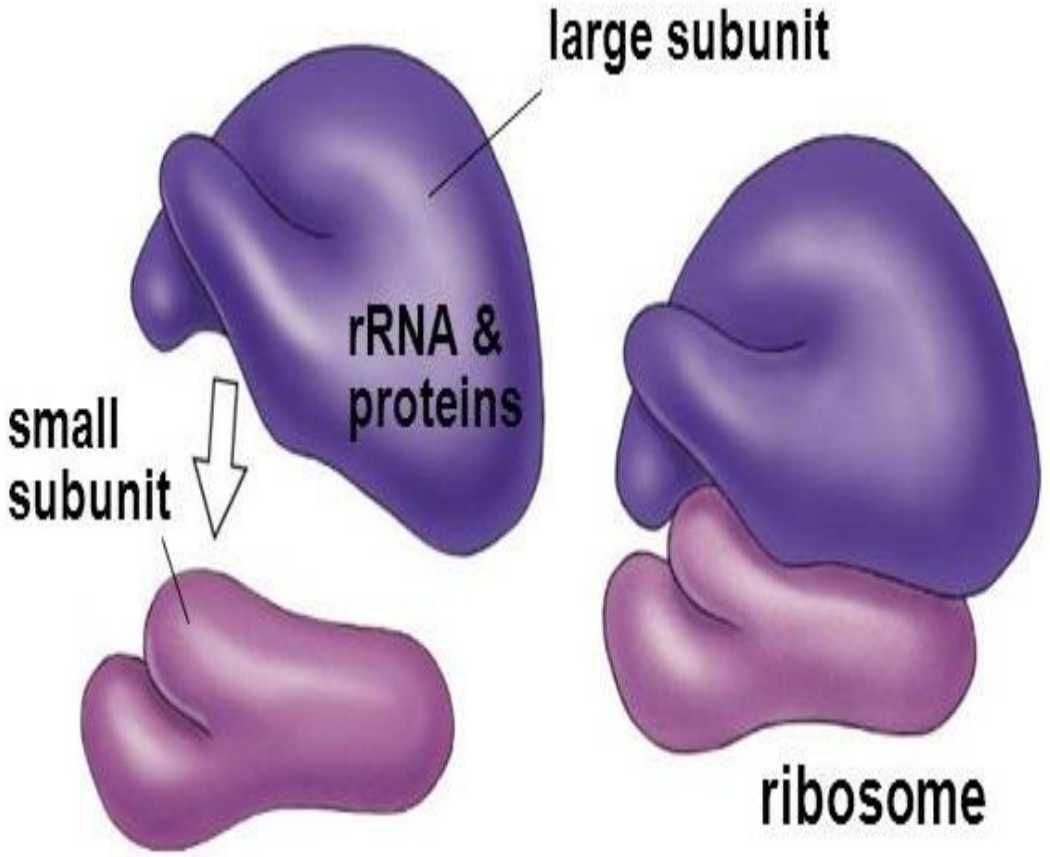
Dr Iman Na

Ribosomes

Dr Imam Mobil

Ribosomes

- Definition: granules of *nucleoproteins* (ribosomal RNA (rRNA) + proteins).
- Site: They are present in all cells *especially* in protein synthesizing cells.
- Structure: *two subunits;* small subunit and large subunit.



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Ribosomes

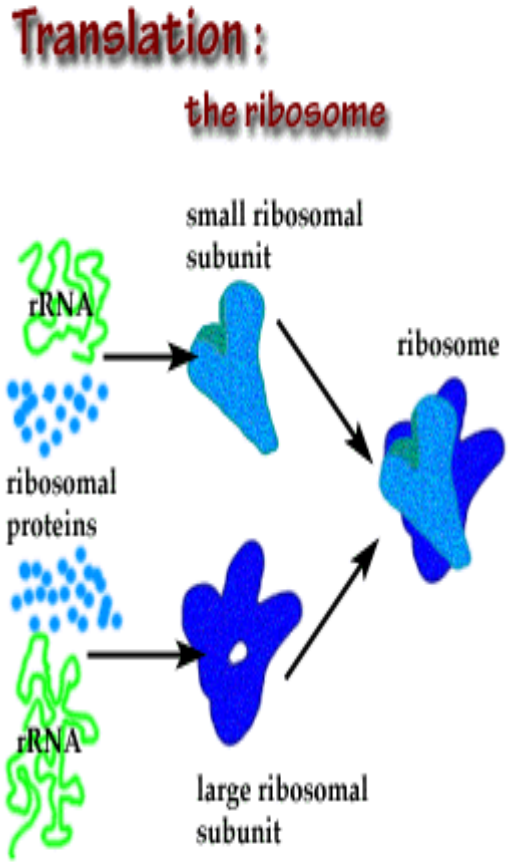
- Synthesis:

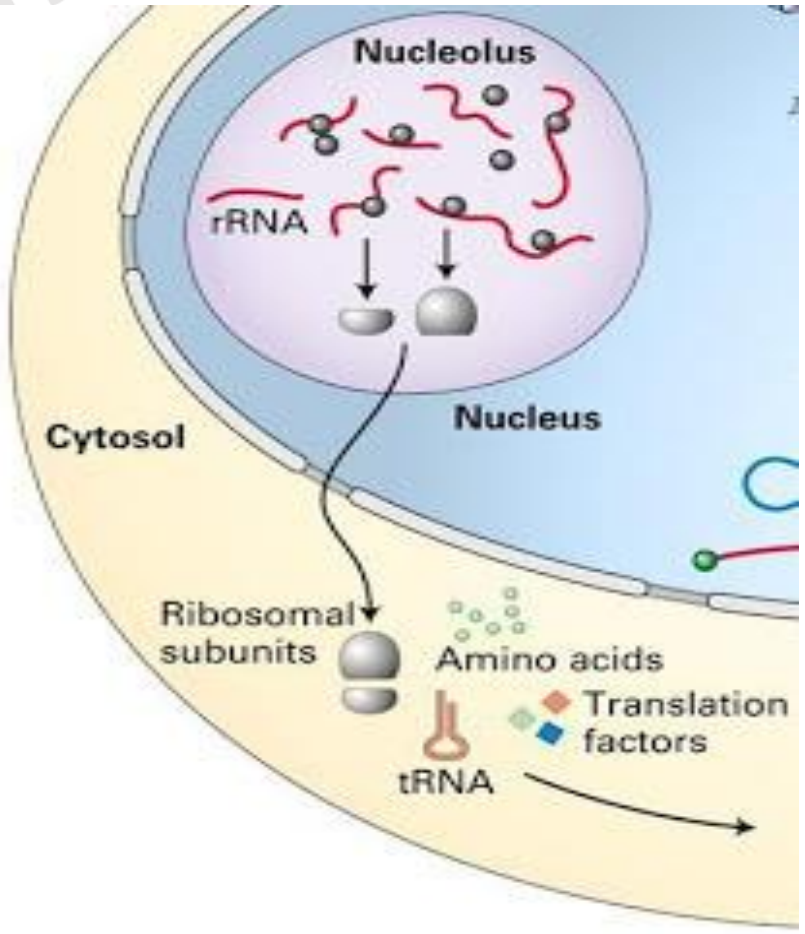
1-The rRNAs are synthesized inside *the nucleolus*.

2-Ribosomal-associated proteins are synthesized in *the cytoplasm*.

3-Ribosomal subunits then leave the nucleus, via the nuclear pores, to enter the cytoplasm.

The small and large subunits are present in the cytosol individually and do not form a ribosome until protein synthesis begins.

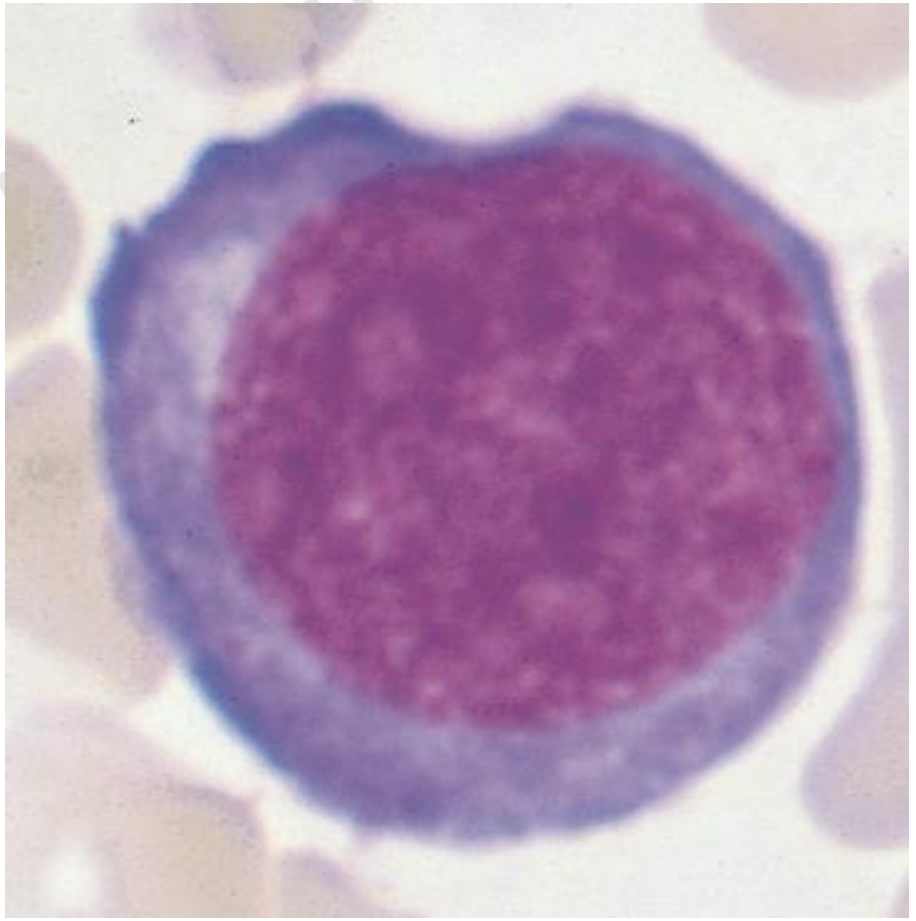




Ribosomes

- LM:

When present in large amounts they cause *cytoplasmic basophilia.*



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Ribosomes

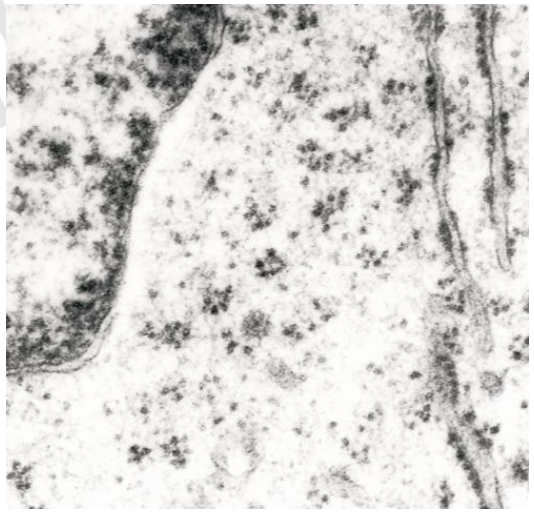
- EM: Ribosomes are small electron dense granules.
- Types:

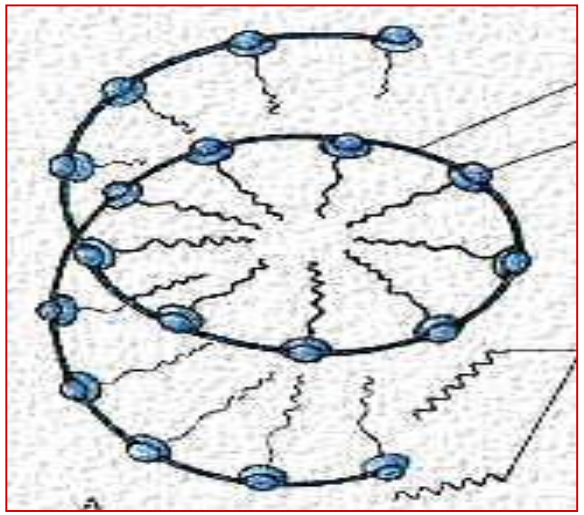
A-Free ribosomes:

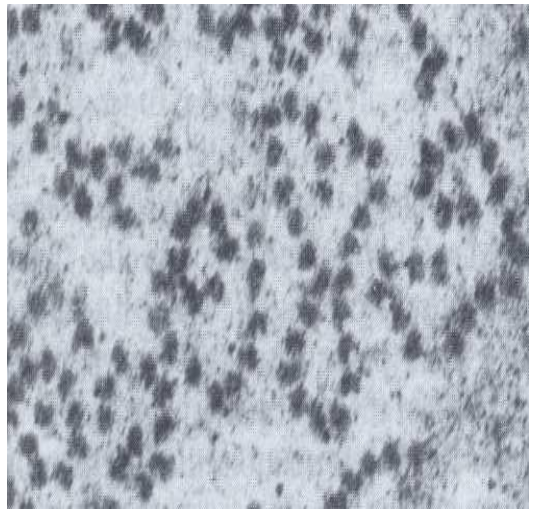
1- Solitary particles: scattered in the cytoplasm.

➢Function: act as *a reserve*.

2- Aggregated (polysomes): clusters of 10 or more connected by single strand of mRNA.



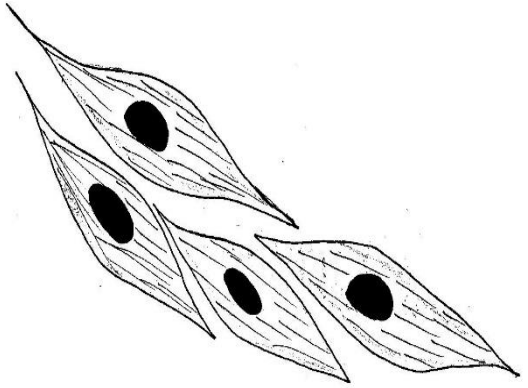


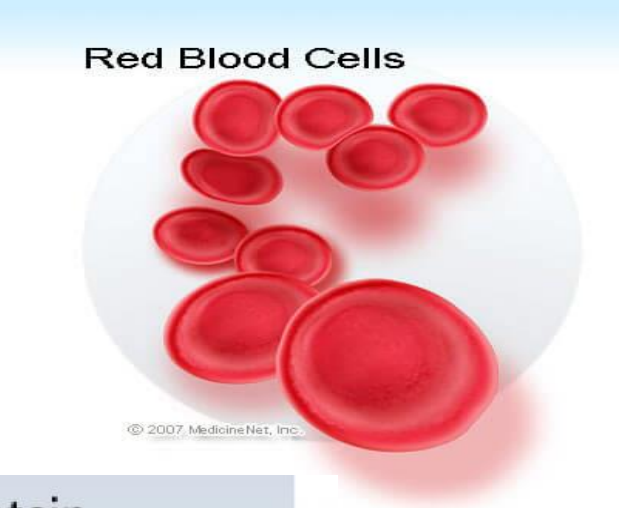


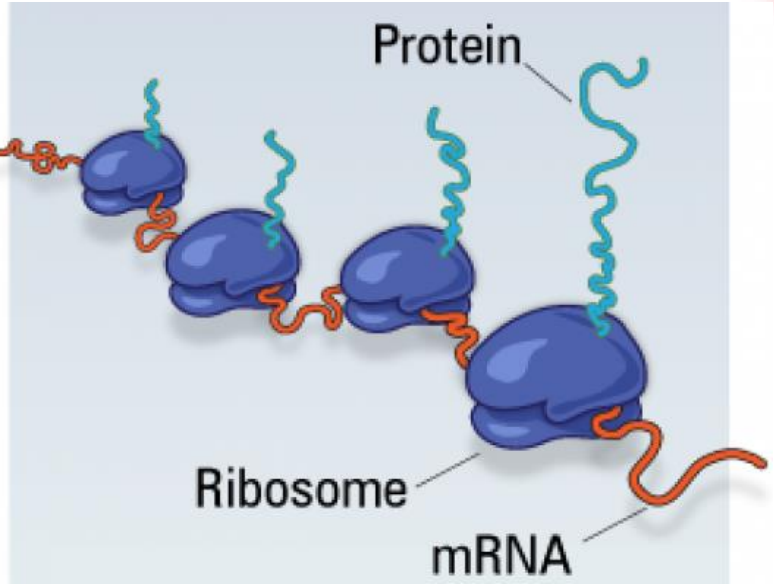
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Ribosomes

- Function of polysomes: responsible for *synthesis of cytosolic protein* e.g. in dividing cells and growing cells, synthesis of hemoglobin in developing red blood cells and contractile protein in muscle cells.

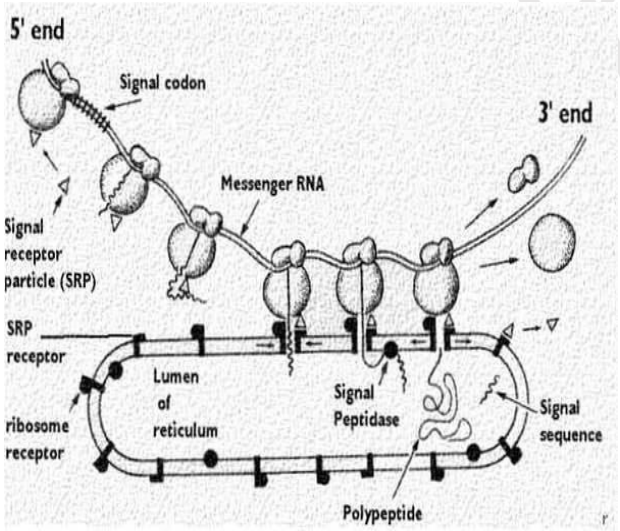


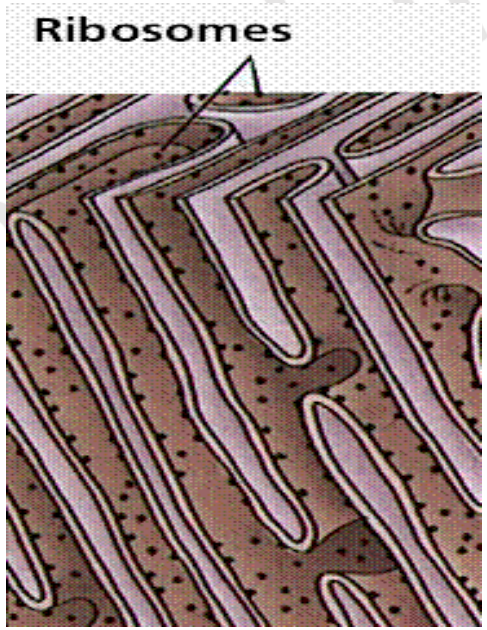


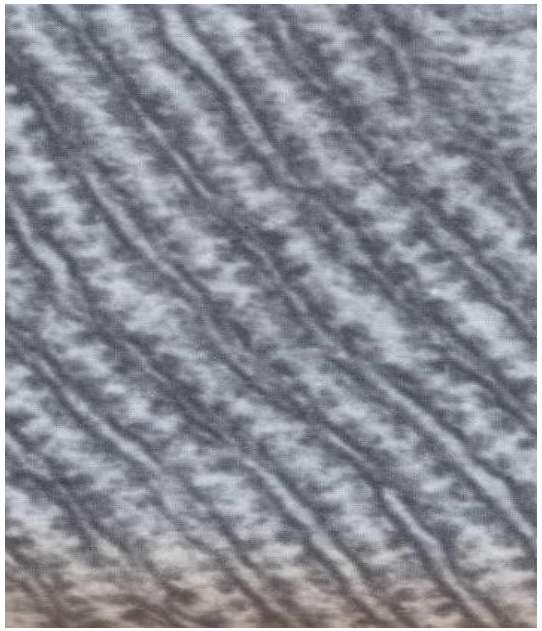


Ribosomes

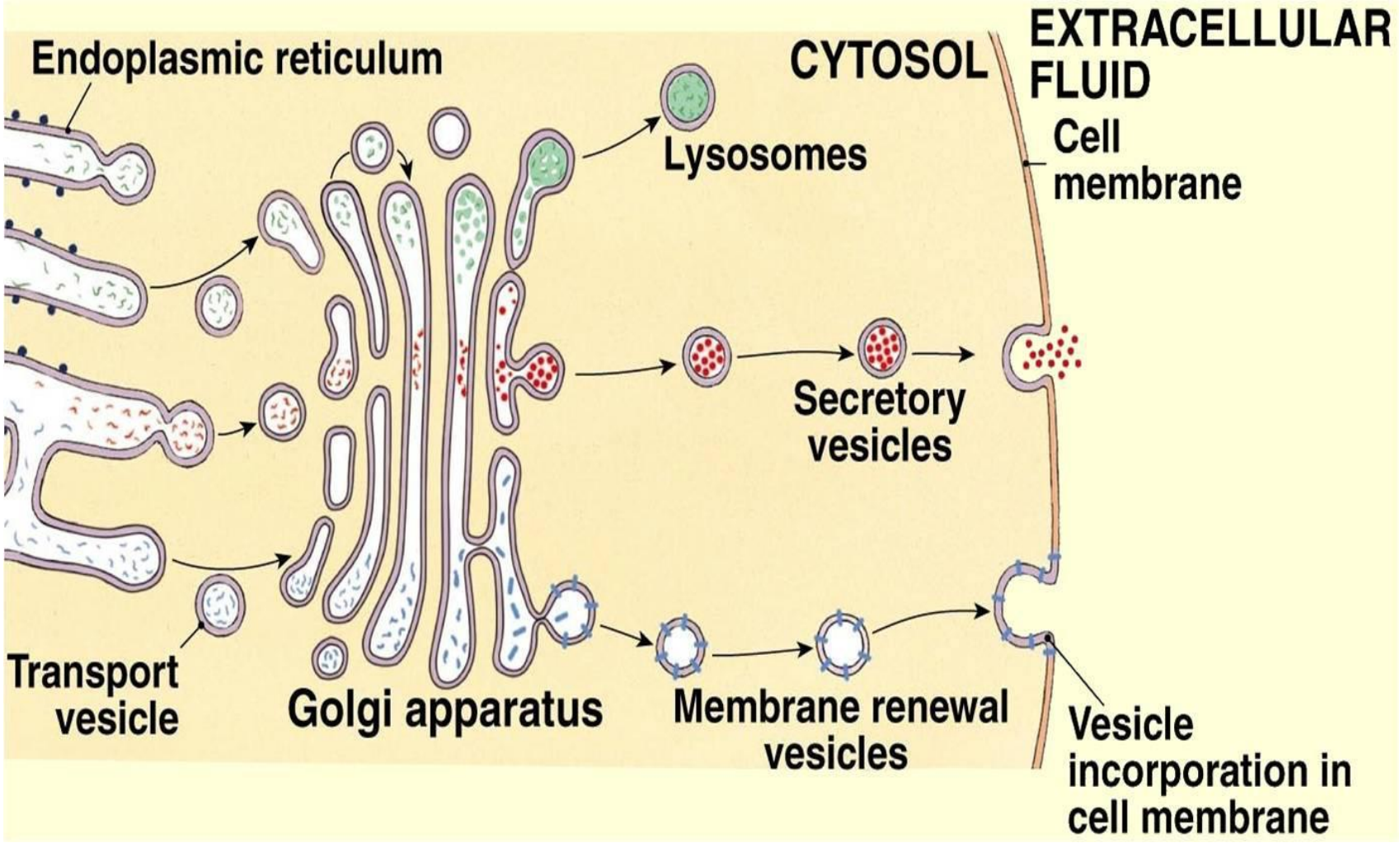
- **B- Attached ribosomes:** these are polysomes that become attached to the outer membrane of the endoplasmic reticulum.





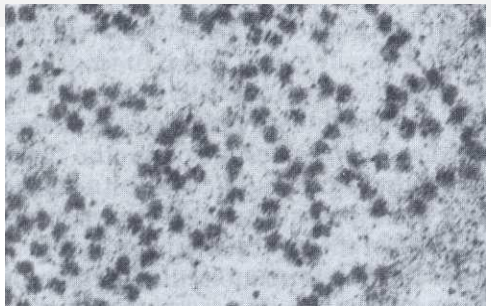


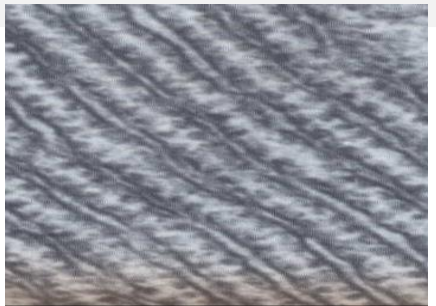
Ribosomes


Endoplasmic reticulum
CYTOSOL
EXTRACELLULAR FLUID
Lysosomes
Cell membrane
Secretory vesicles
Transport vesicle
Golgi apparatus
Membrane renewal vesicles
Vesicle incorporation in cell membrane

Types of ribosomes	Free solitary ribosomes	Free aggregated ribosomes (polysomes)	Attached ribosomes
1- LM		Not seen but in large amount give cytoplasmic basophilia.	Not seen but in large amount give cytoplasmic basophilia.
2- EM	Small electron dense particles.	10 or more ribosomes connected by a single strand of mRNA.	Small electron dense particles attached to rER.
3- Function	Reserve.	Synthesis of cytosolic proteins (used within the cell)	Synthesis of secretory proteins, lysosomal enzymes& membrane proteins.







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Endoplasmic reticulum

Dr Imaan Ibil

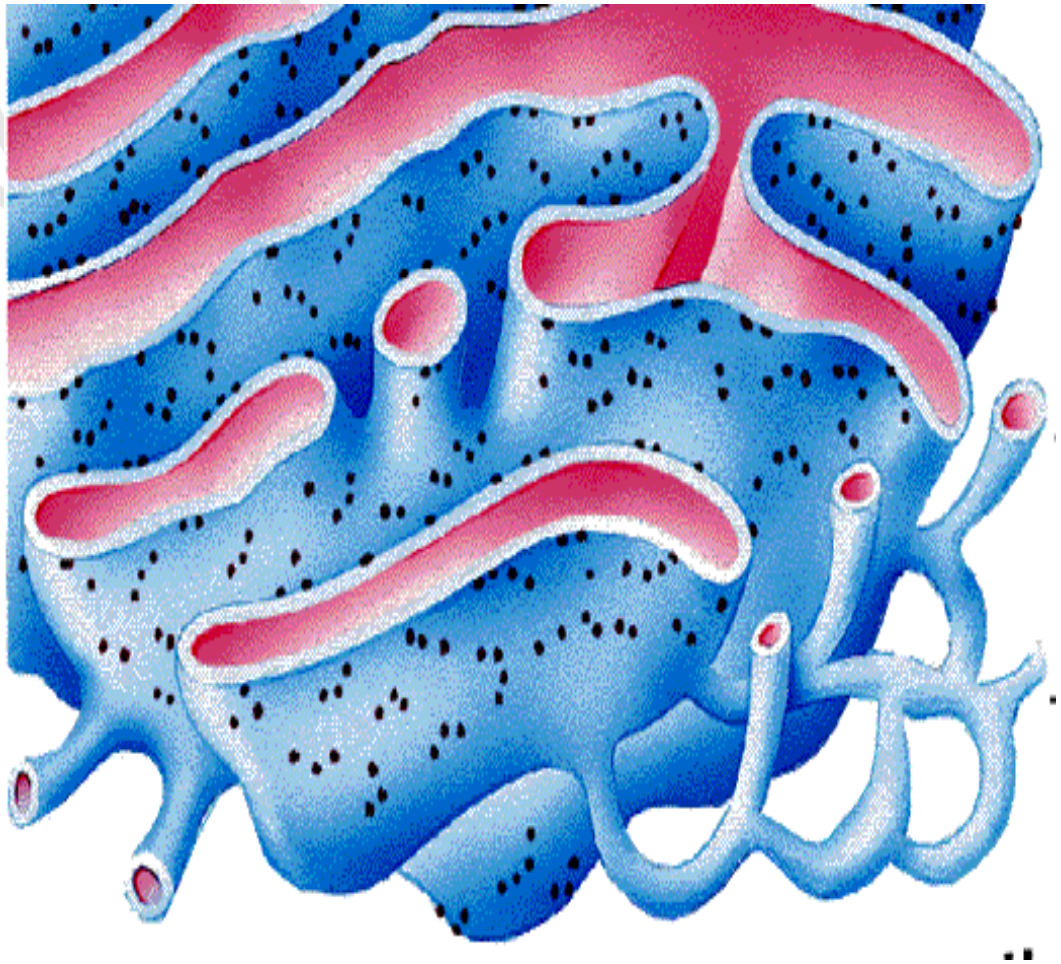
Endoplasmic reticulum

- The endoplasmic reticulum forms the most extensive membrane system in the cytoplasm.
- *The ER has two types:*

1-Rough endoplasmic reticulum (rER).

2-Smooth endoplasmic reticulum (sER).

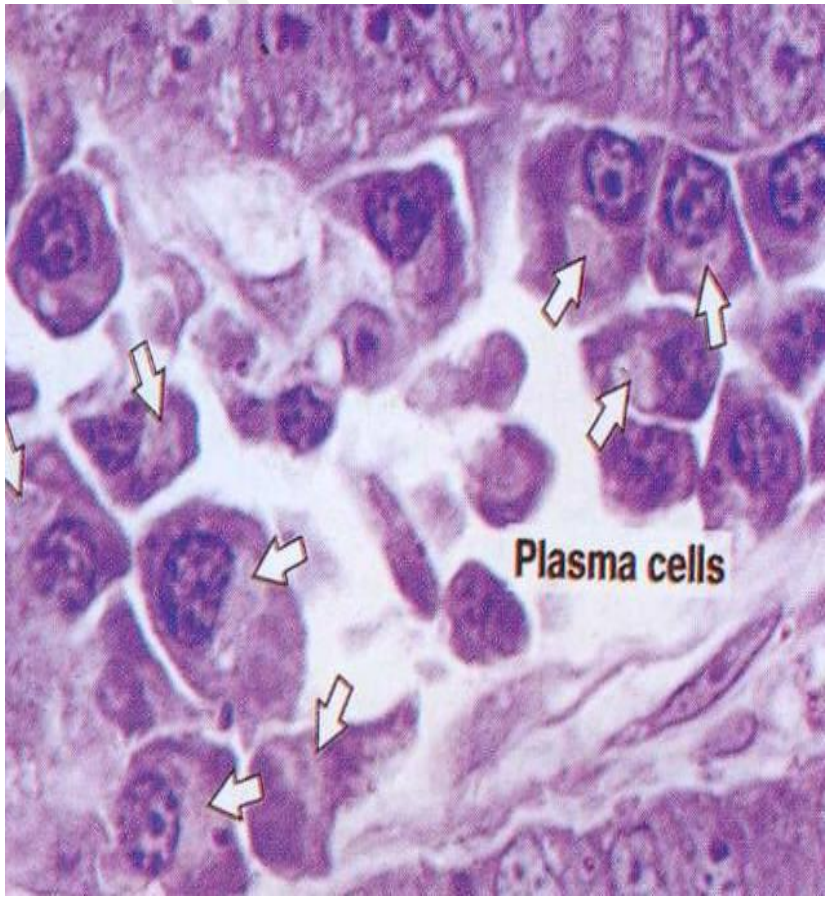
- *Both types form a single membrane system*



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Rough endoplasmic reticulum (rER)

- **LM :**When present in large amounts they cause *cytoplasmic basophilia* due to their attached ribosomes.



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Rough endoplasmic reticulum (rER)

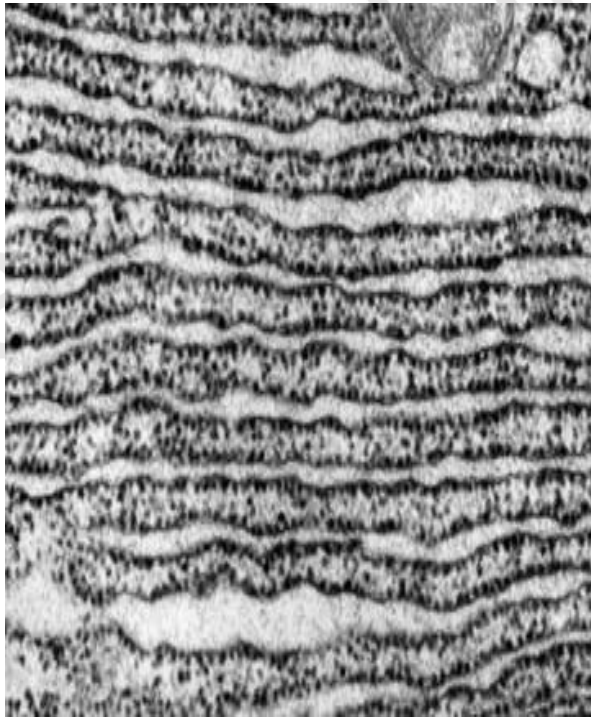
- **EM :**

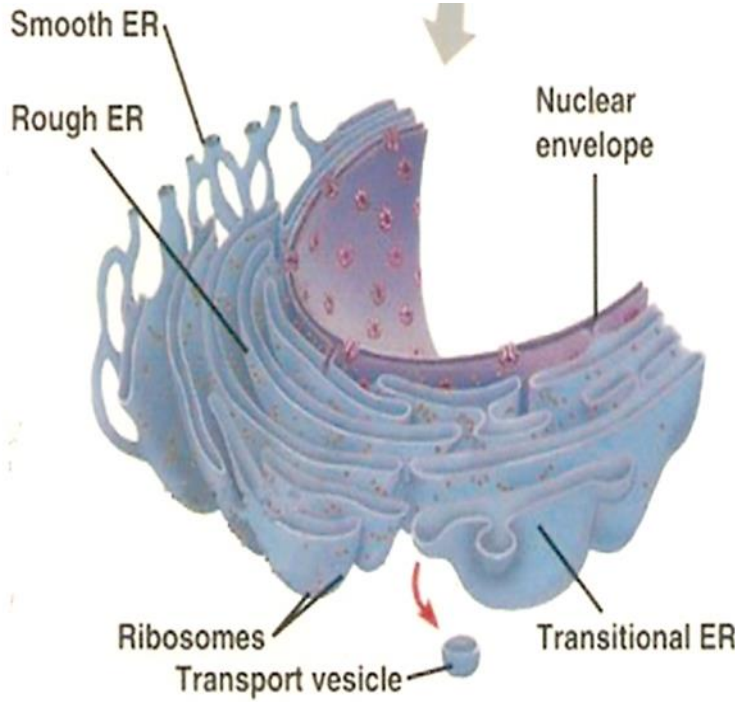
1-It consists of interconnected parallel flattened sacs called *cisternae*.

2- Its outer surface is studded *with ribosomes.*.

3-It is continuous with the outer membrane of the nuclear envelope.

4-The lumen contains flocculent material that represents newly-formed protein.





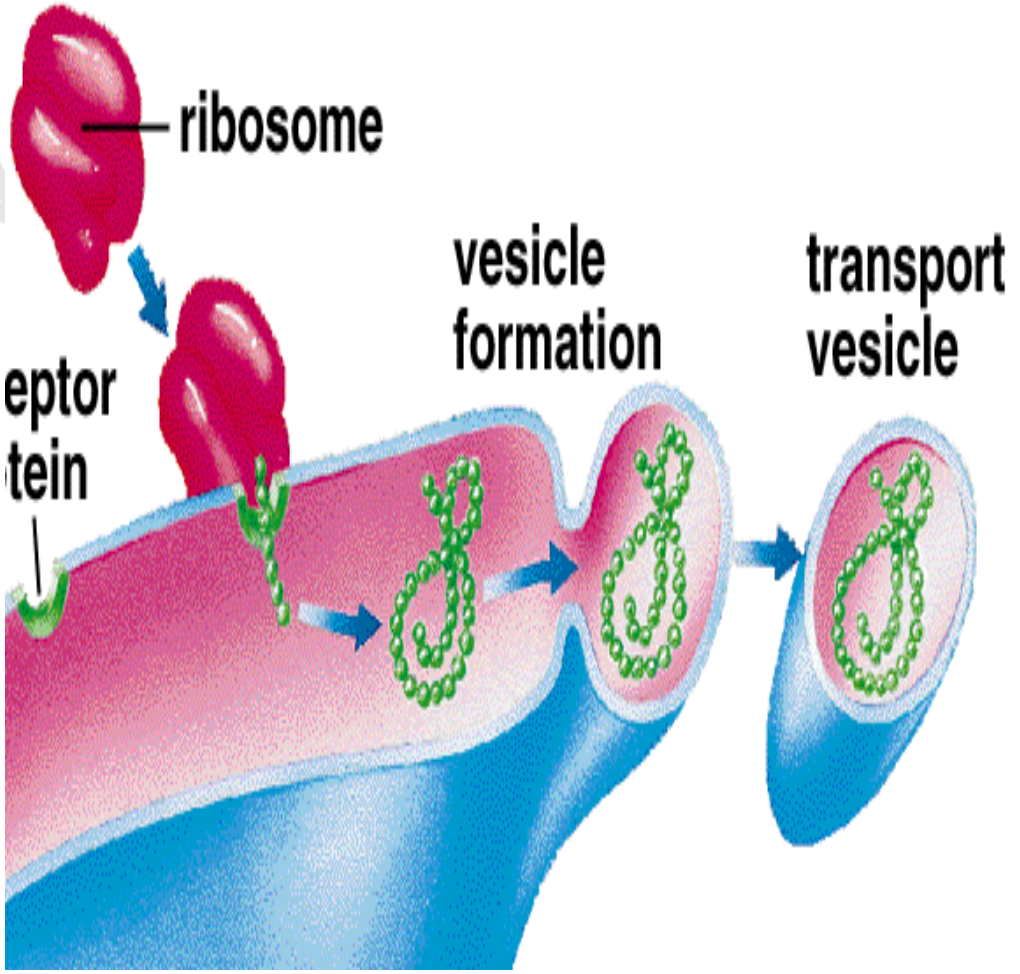
Rough endoplasmic reticulum (rER)

- **Functions:**

1- *Synthesis of* secretory proteins, lysosomal enzymes and proteins inserted into the cytoplasmic membranes.

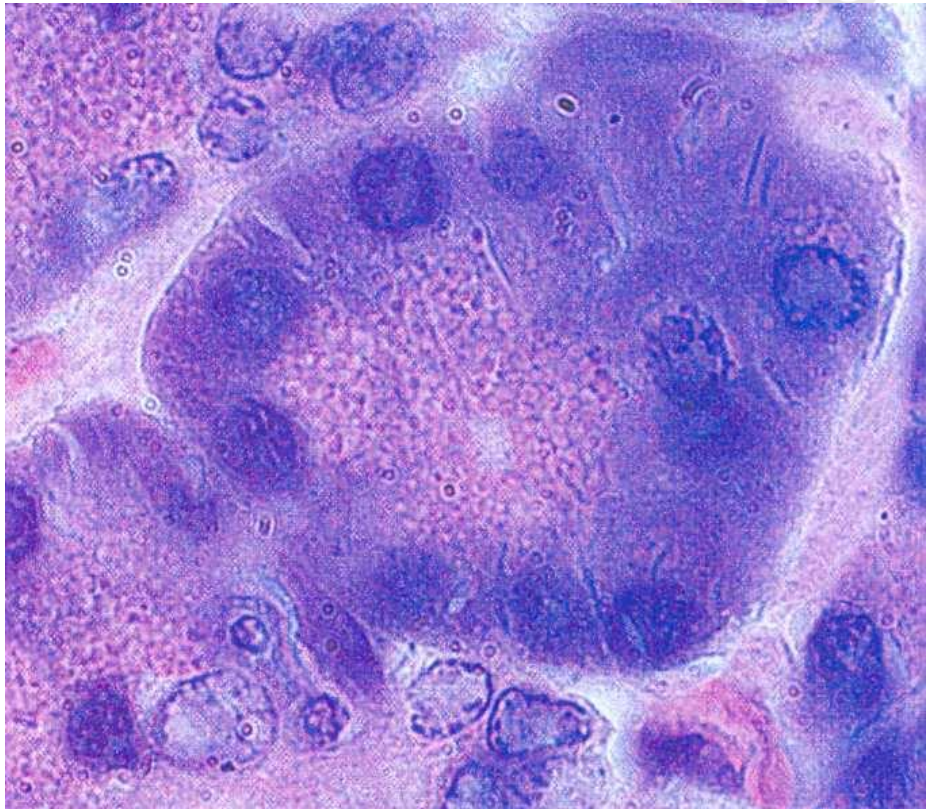
2- *Post-translational modification* of the newly-formed protein e.g.folding, sulfation and initial glycosylation.

3- *Transport* the newly-synthesized protein to the Golgi body by transport vesicles.



Rough endoplasmic reticulum (rER)

- Sites: Protein synthesizing and secreting cells e.g. liver cells, pancreatic acini, fibroblasts and plasma cells.

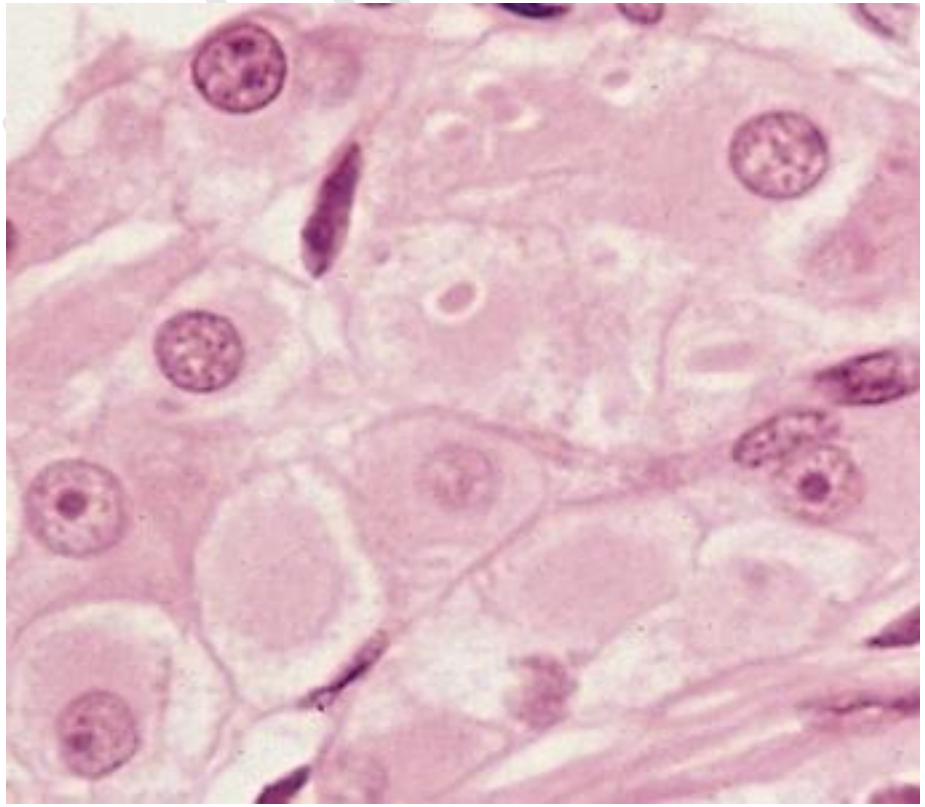


Nabil

Smooth endoplasmic reticulum (sER)

- **LM :**

Cells with large amounts of sER exhibit *cytoplasmic eosinophilia*.



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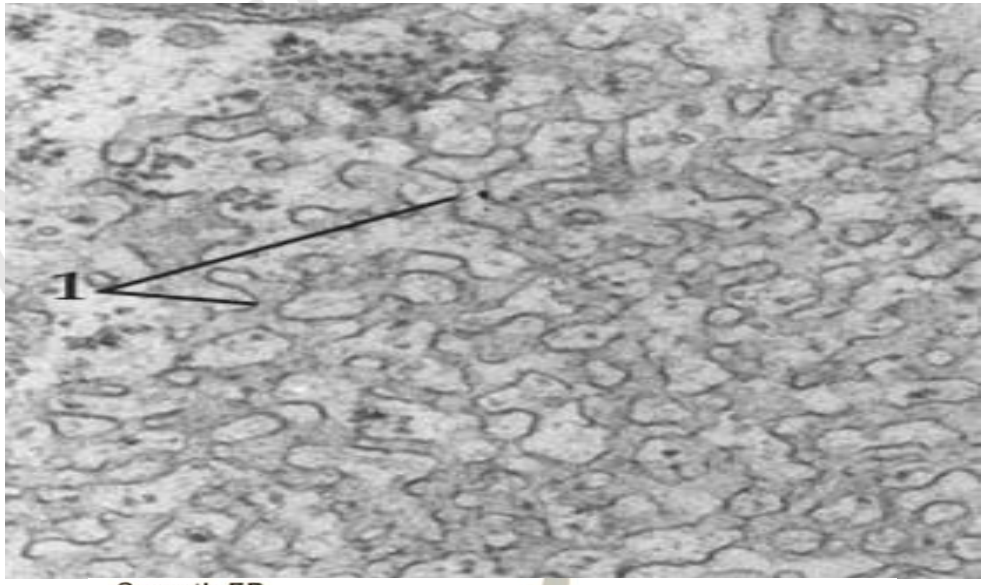
Smooth endoplasmic reticulum (sER)

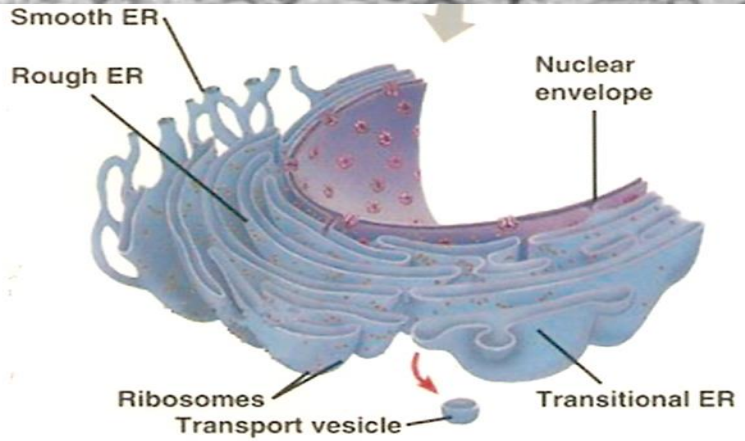
- **EM :**

1-It consists of close network of *interconnected branching tubules and vesicles*.

2-The membranes have smooth surface.

3-The membranes of the sER are continuous with that of rER.





Smooth endoplasmic reticulum (sER)

- **Functions:**

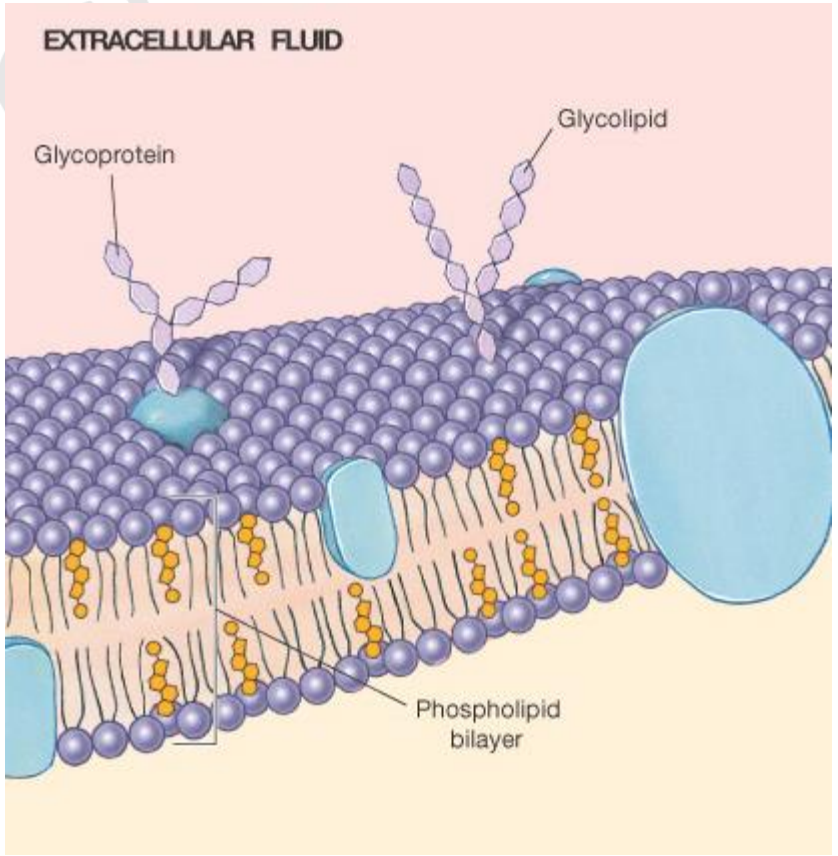
1-Synthesis of *membrane lipids*; the phospholipids and cholesterol.

2-Synthesis of *steroid hormones*.

3-Synthesis of *glycogen* in liver.

4-*Detoxification* of toxic substances e.g. alcohol and drugs.

5-Regulation of *calcium ions* during muscle contraction.

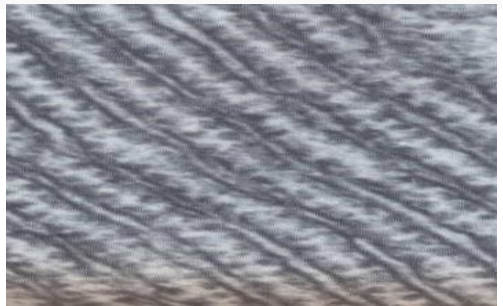


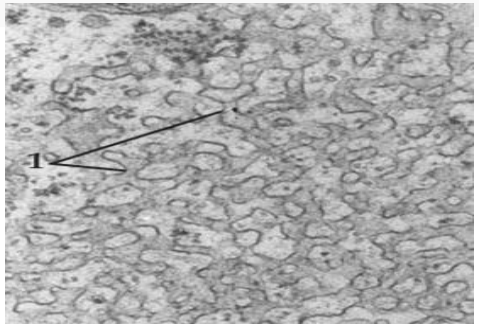
Sites: steroid-secreting cells (in the adrenal cortex, testis and ovary), liver cells & muscles.

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Types of endoplasmic reticulum	Rough endoplasmic reticulum	Smooth endoplasmic reticulum
1- LM	Not seen but in large amount give cytoplasmic basophilia.	Not seen but in large amount give cytoplasmic acidophilia.
2- EM	Parallel, flattened interconnected tubules. Studded with ribosomes.	Interconnected branching tubules and vesicles. No ribosomes.
3- Functions	1- Synthesis of secretory proteins, lysosomal enzymes& membrane proteins. 2- Post translational modification of protein. 3- Transport protein to Golgi	1- Synthesis of lipid & cholesterol of the cell membrane. 2- Synthesis of steroid hormones. 3- Synthesis of glycogen. 4- Detoxification of toxic substances. 5- Storage of calcium in muscles.
4- Sites	Protein secreting cells e.g. liver , fibroblasts.	Steroid secreting cells, liver & muscles.

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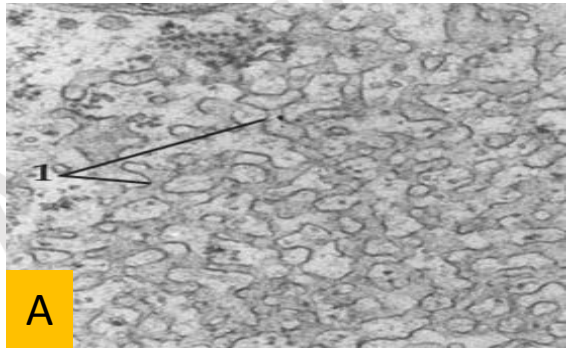


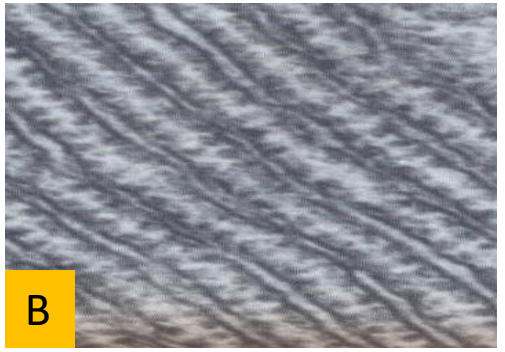
Which diagram represents the site of synthesis of contractile proteins of the muscle fibers?

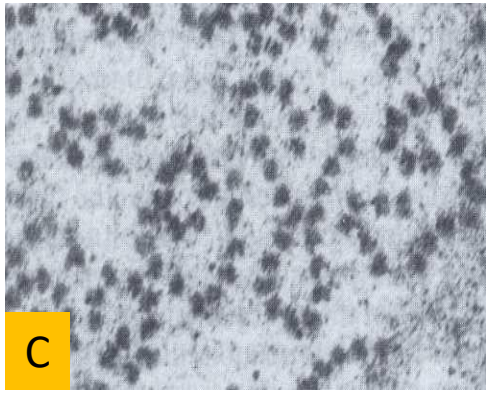
1-A.

2-B.

3-C. Polysomes







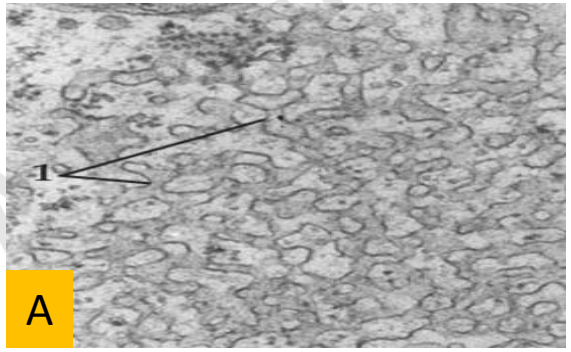
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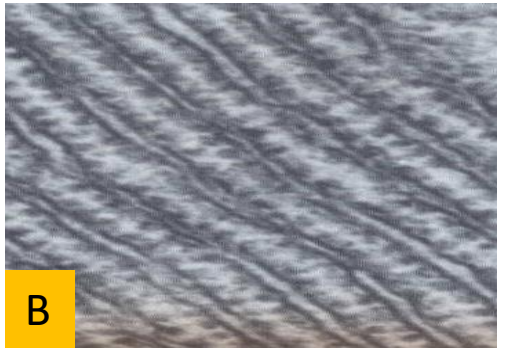
Which diagram represents the site of synthesis of contractile proteins of the muscle fibers?

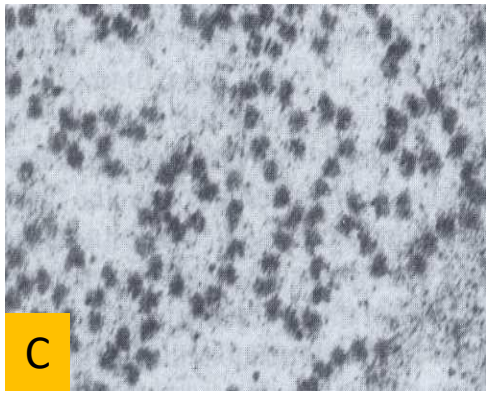
1-A.

2-B.

3-C. ✓







Dr Iman


THANK
YOU


Thank You